

When Will A Power Outage Occur?

Goal Statement

Be able to predict time of day of power outages based on factors about the event affecting power across the United States with a mean value error of less than 30 minutes.

Research Question:

Can we develop a model that can predict the time of day of U.S. power outages in 2023 caused by various events?

Modeling Approach

For our modeling approach we will compare three different models to see which can best forecast the time of day of power outages in the U.S. [3]. We will start with a linear regression model, which will serve as our baseline model. We will then run a random forest model and gradient boosting model as the other two models in our comparison. We will then compare the models using various evaluation metrics. Our main evaluation metric will be the mean value error (MAE), but we will also look at the R^2 score and the root mean squared error (RMSE).

Executive Summary

This document outlines our dataset establishment details and analysis plan for forecasting power outages using time-series modeling.

Dataset Establishment Details

Our data is gathered from a dataset on Data.gov called Event-correlated Outage Dataset in America. The dataset in Data.gov contained two different zip files, with several csv files within each zip. The zip file we chose to work with is called eaglei_outages_with_events_2023 which contains roughly 77,000 rows of data. The dataset was very clean to begin with so the only cleaning we did was drop unnecessary columns (not included in the data dictionary). We also added a new column which converted the start_time column into the hour of the day which the outage began.

Data Dictionary:

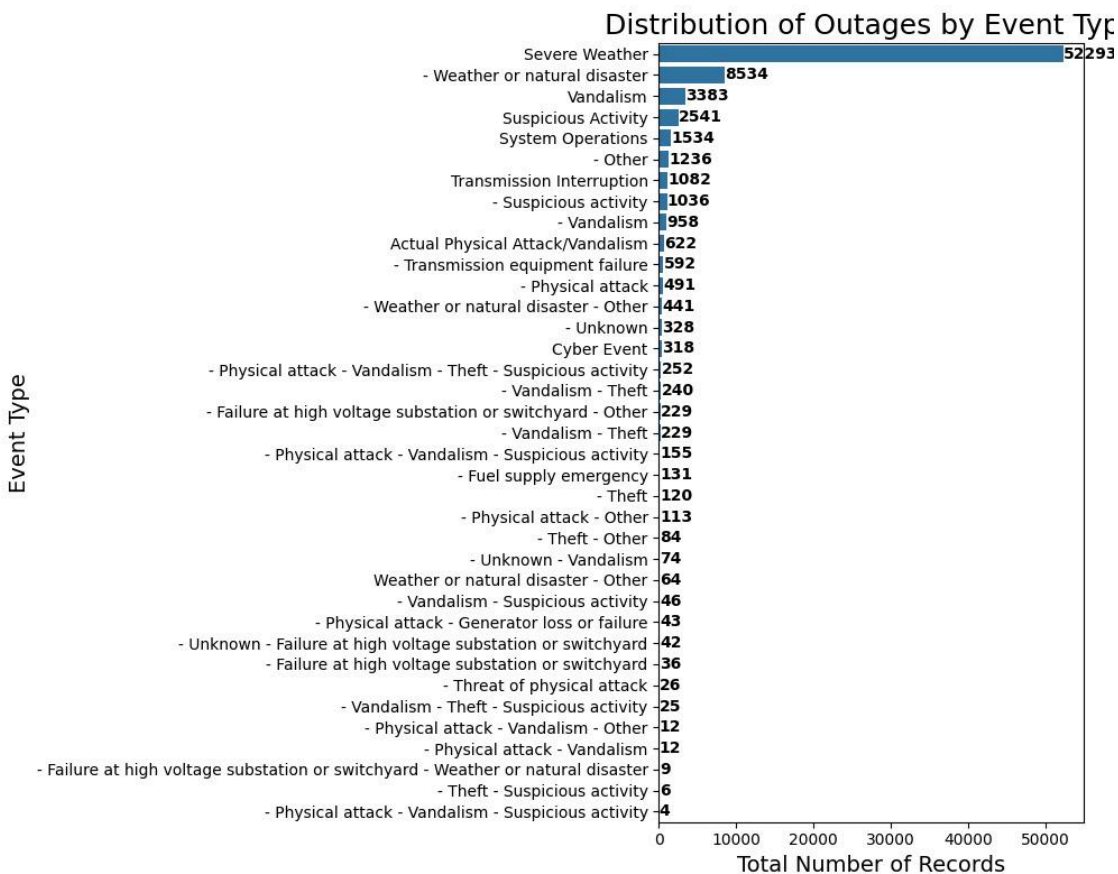
Column	Description
event_id	The count of outage in each state
Event Type	Cause of event
state	State in which outage occurs
county	County (in each state) in which outage occurs
start_time	The start time of the outage
duration	The length of time of the outage

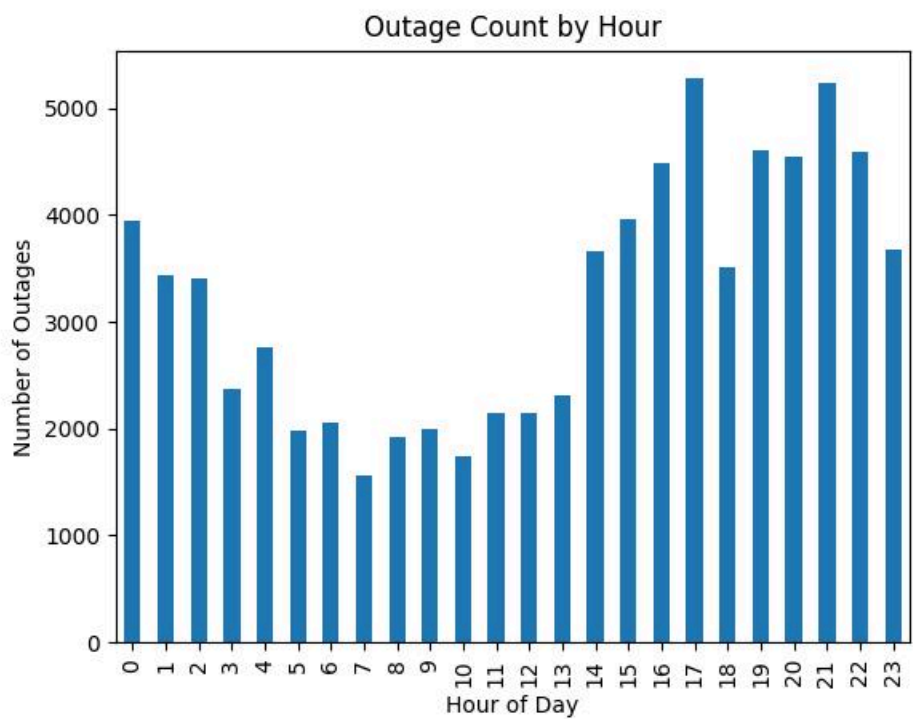
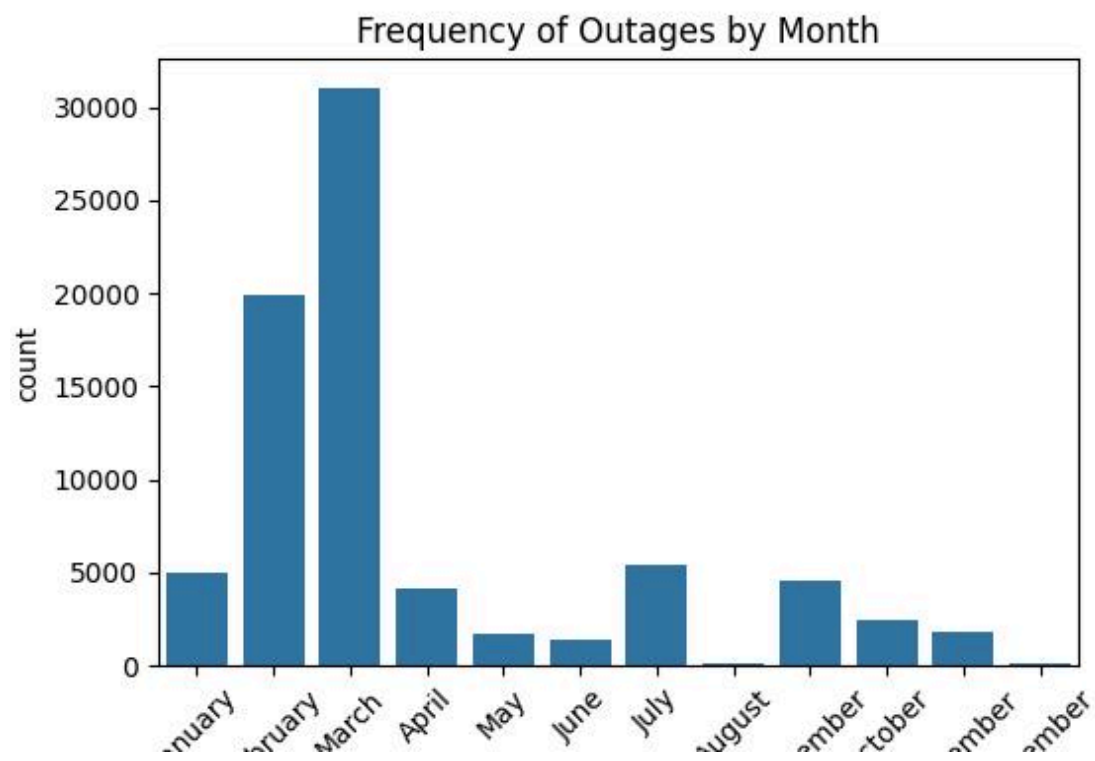
end_time	The end time of the outage
hour	The time of day an outage began

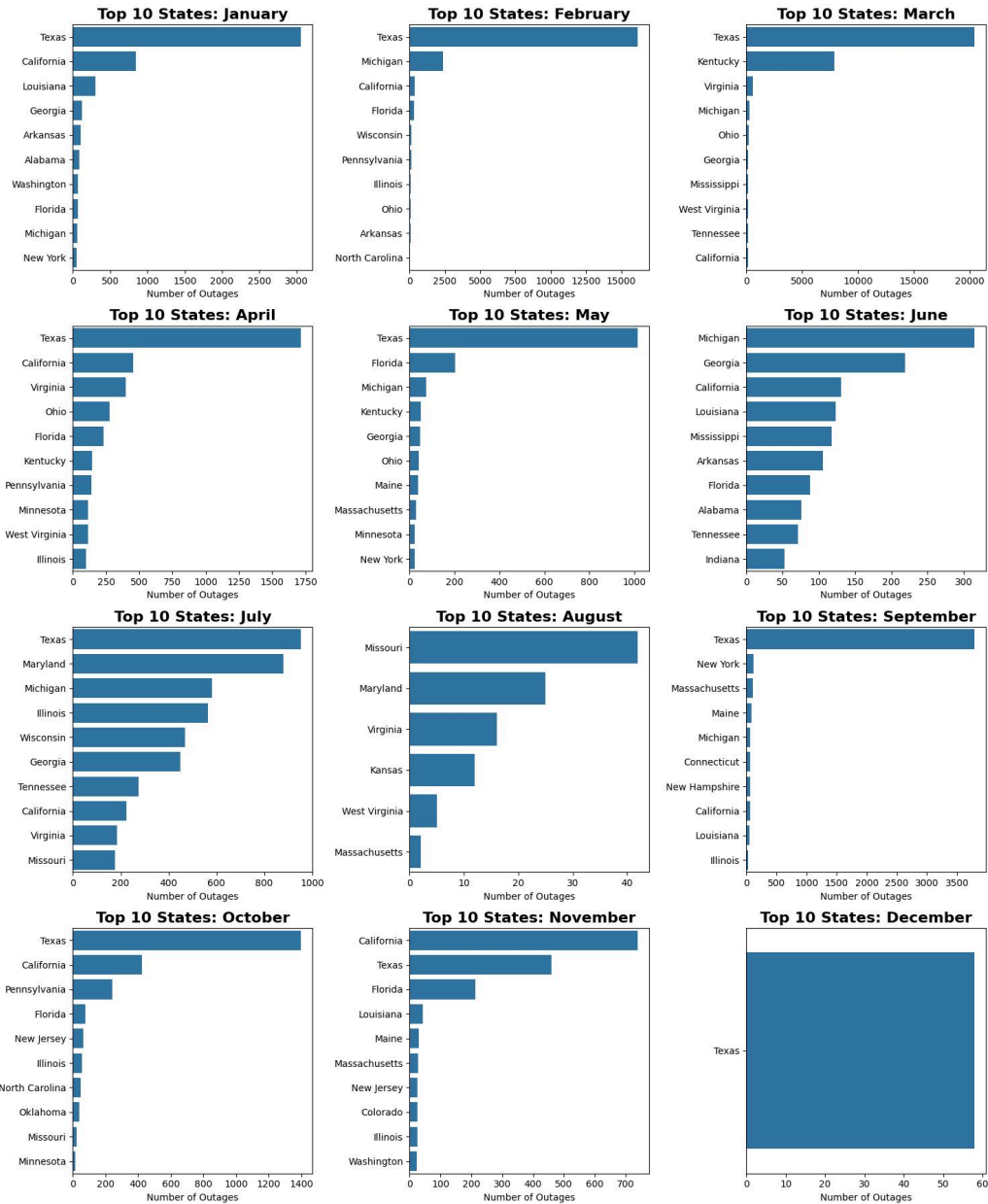
We explored the distribution of the dataset by event type and found that most of the events were categorized as severe weather. We also explored the frequency of outages by month and found that March had the highest frequency of outages. We summed the mean customers for each event in each state and found that the state with the highest number of customers is Texas, by a significantly large margin. We also looked at the top ten states by frequency of outage per month and found Texas to be in the top ten for ten months of the year (minus June and August).

To fit with a more time-series approach to the question we changed our original research question of looking at the duration of power outages to trying to predict at what time in the day they will occur which led to changes in our code which have already been addressed and appropriately documented in our analysis plan.

This brings us to the question of: Can we create a model that can effectively predict what time of day a power outage will occur?







Analysis Plan

Step 1: Find Data and Preprocessing

- Search through approved data source locations to find a data set with respect to time series data. Once final selection is made, begin preprocessing preprocessing. We cleaned the data as discussed in the [Dataset Establishment Details](#) section.

Step 2: Data Analysis

- We conducted EDA to understand the structure and characteristics of the dataset. This included examining the top ten states affected by outages each month, the frequency of

outages by month, the distribution of outages by event_type, and the distribution of outages by hour. In order to do this we converted the start_time and end_time columns to datetime objects that could be used in EDA and for modeling. We also extracted useful features such as month and hour, that could be used for plots in EDA. We also set the month order so that the EDA would appear chronologically.

Step 3: Training Strategy and Data Splitting

- Before running any models, the data needs to be split. Before splitting the data, the categorical variables need to be converted to one-hot encoded columns that can be used as numerical variables in the model. These will serve as the feature matrix (X variable). We will define the y-variable as the target variable which will be hour. Then the data will be split into a test and train set. The train set will then be scaled and then converted back to DataFrames. The data is then ready to be used in the models.

Step 4: Model Development (Linear Regression, Gradient Boosting, and Random Forest)

- We will run three models: a Logistic Regression as a baseline, a Random Forest, and a Gradient Boosting. In order to do this. We will run these models without hyperparameter tuning to serve as an initial performance of the model and then we will run the models again with added hyperparameter tuning to see if that affects model performance.

Step 5: Model Evaluation

- The models will be evaluated via R^2 score, mean absolute error, and root square mean error, which explain the variance of the model and how many hours the model is off by.

Step 6: Comparative Analysis

- We will conclude by comparing the models evaluation metrics and determining which of the three models is the most effective in predicting what time of day an outage will occur. We will know if our model is successful if they are able to predict the time of day a power outage will occur with a mean value error of less than 30 minutes.

References

[1] “Emergency electrician response times: For different situations,” ShockFree Electrical, <https://shockfreeelectrical.com/blog/emergency-electrician-response-times-for-different-situations/> (accessed Feb. 25, 2026).

[2] “Data.gov,” *Data.gov*, 2025. <https://catalog.data.gov/dataset/event-correlated-outage-dataset-in-america> (accessed Feb 25, 2026).

[3] M. Mohan, T. Ayyalasomayajula, Santhosh Bussa, and S. Ayyalasomayajula, “Forecasting Home Prices Employing Machine Learning Algorithms: XGBoost, Random Forest, and Linear Regression,” *Journal of Engineering & Technology Advancements*, vol. 1, no. 1, pp. 2583–2646, doi: <https://doi.org/10.56472/25832646/JETA-V1I1P114>.